Physically Plausible Human–Object Rendering From Sparse Views via 3D Gaussian Splatting

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***Abstract*—Rendering realistic human-object interactions (HOIs) from sparse-view inputs is a challenging yet crucial task for various real-world applications. Existing methods often struggle to simultaneously achieve high rendering quality, physical plausibility, and computational efficiency. To address these limitations, we propose HOGS (Human-Object Rendering via 3D Gaussian Splatting), a novel framework for efficient HOI rendering with physically plausible geometric constraints from sparse views. HOGS represents both humans and objects as dynamic 3D Gaussians. Central to HOGS is a novel optimization process that operates directly on these Gaussians to enforce geometric consistency (i.e., preventing inter-penetration or floating contacts) to achieve physical plausibility. To support this core optimization under sparse-view ambiguity, our framework incorporates two pre-trained modules: an optimization-guided Human Pose Refiner for robust estimation under sparse-view occlusions, and a Human-Object Contact Predictor that efficiently identifies interaction regions to guide our novel contact and separation losses. Extensive experiments on both human-object and hand-object interaction datasets demonstrate that HOGS achieves state-of-the-art rendering quality and maintains high computational efficiency.**

***Index Terms*—Human-object interactions, 3D Gaussian splatting, sparse-view rendering, physically plausible optimization.**

I. Introduction

RENDERING dynamic real-world scenes is crucial for applications ranging from immersive media to robotic interaction [1], [2], [3], [4], [5], [6]. A fundamental challenge in deploying these applications, however, is the reliance on sparse-view inputs [7], [8], [9] (6 or fewer fixed camera views covering a 360° circle), as dense multi-view camera setups are often impractical in real-world environments. This limitation becomes particularly pronounced when rendering human-object interactions (HOIs), where the standard for realism extends beyond simple visual fidelity [10], [11], [12], [13]. For these specific scenes, achieving physical plausibility is paramount. Consider the need to verify that a user's avatar is sitting correctly on a virtual chair rather than sinking into it for an immersive AR experience, or to confirm a driver's hand is making proper contact with the steering wheel for a safety analysis. Such scenarios demand a level of physical realism that general-purpose renderers struggle to provide, especially given the inherent ambiguity of sparse-view inputs. Therefore, jointly achieving high-quality rendering and robust physical plausibility for HOIs from sparse views remains a pivotal and challenging goal.

Meanwhile, HOI rendering has witnessed significant progress within recent years [14], [15], [16], [17]. Early approaches rely on 3D mesh reconstruction combined with per-frame texture mapping [18], [19], [20], often incorporating physical optimizations to enforce plausible contact between the reconstructed human and object meshes (*i.e.*, ***physically plausible***). While they provide basic visual fidelity, they are susceptible to occlusions and incomplete textures. These challenges are further exacerbated in sparse-view settings, which limit their effectiveness in real scenarios. In light of these issues, more advanced rendering techniques are proposed to enhance HOI rendering, among which neural rendering techniques [15], [21], [22], [23] demonstrate significant gains in ***rendering quality***. A common strategy for neural HOI rendering uses a layer-wise NeRF pipeline to represent and render both human and object [14], [24], enabling free-viewpoint rendering. Despite the impressive visual fidelity achievable, this type of approach inherently demands dense multi-view inputs and incurs substantial computational overhead, limiting its practicality for real-time applications. Moreover, even with the adoption of layered representations to disentangle human and object, these pipelines generally lack dedicated mechanisms to ensure physically plausible interactions in intricate HOI scenarios.

To tackle the computational limitations of NeRF-based methods, recent research has explored 3D Gaussian Splatting (3DGS) [25] as a powerful alternative, offering remarkable ***rendering efficiency*** for photo-realistic rendering of both static and dynamic scenes [26], [27], [28], [29]. This explicit Gaussian representation provides robustness to sparse views by efficiently integrating limited information and minimizing

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reconstruction ambiguity [30], [31], [32], [33]. Building upon these strengths, 3DGS has been applied to dynamic human modeling, achieving high-quality reconstruction of animatable human avatars [34], [35], [36], [37], [38]. Despite these advancements, existing 3DGS-based methods predominantly focus on single-human rendering. Applying them directly to complex interactions between humans and objects reveals specific limitations due to severe occlusions and geometric ambiguity.

In this paper, we propose **HOGS** (**H**uman-**O**bject Rendering via 3D **G**aussian **S**platting), a novel framework explicitly designed to address the specific challenges in sparse-view HOI rendering. The rationale behind our framework design follows a logical progression to address the limitations of standard 3DGS in this domain:

First, in our sparse-view setting, geometric information is severely missing. Without strong priors, standard 3DGS tends to produce "floaters" or fails to converge accurately. To address this, we introduce a **Human-Object Deformation** module that leverages parametric priors (SMPL-H [39] and templates) to provide a robust geometric initialization. This step is a prerequisite for 3DGS to work under sparse views, ensuring the optimization starts from a plausible geometric foundation.

Second, following deformation, we obtain initialized points from two fundamentally different sources: the SMPL-based human (articulated) and the template-based object (rigid). Treating them as a single homogeneous Gaussian cloud is problematic because they require distinct densification and pruning strategies due to their differing dynamic properties. Conversely, optimizing them in complete isolation fails to model the correct visibility and occlusion relationships. To resolve this, we design the **Composed Gaussian Splatting** strategy. We maintain separate Gaussian sets to apply entity-specific learning rates and densification rules, but critically, we merge them into a unified scene for rasterization. This strategy ensures that: 1) each entity is optimized with appropriate dynamics; and 2) the relative depth and occlusion are correctly learned via the unified 2D projection.

Third, while Composed Gaussian Splatting reconstructs overall visual appearance, it relies primarily on 2D joint human-object mask supervision. Under sparse views, this 2D supervision is often insufficient to constrain the 3D geometry in contact regions, leaving the model blind to the precise physical interface. This leads to visually plausible but physically impossible artifacts, such as inter-penetration or floating contacts. To bridge this gap, we introduce **Physically Plausible Optimization**, which explicitly injects 3D physically plausible geometric constraints that 2D images cannot provide. Specifically, we design a Contact Loss to eliminate unnatural gaps and a Separation Loss to resolve inter-penetration, serving as the final regularizer to ensure the interaction is not just visually clear, but physically valid.

As illustrated in Figure 1, HOGS addresses the limitations of existing methods by combining this efficient joint rendering of human and object Gaussians with a physically plausible optimization process. Through this holistic design, HOGS simultaneously achieves high rendering efficiency, superior visual quality, and geometric plausibility—a harmony that previous approaches have not attained.

Our main contributions are summarized as follows:

- We present HOGS, a novel framework for sparse-view HOI rendering. Our method achieves a significant performance leap, surpassing state-of-the-art 3DGS-based methods by 3.4 dB in PSNR while maintaining real-time rendering speeds.
- We propose a novel optimization mechanism for physical plausibility based on geometric consistency, which constitutes the core of our method. Tailored for unstructured 3D Gaussians, this mechanism integrates SDF-based constraints to explicitly enforce non-penetration and accurate contacts.
- To support this core pipeline under sparse-view ambiguity, we develop a suite of auxiliary components, including a strategy for robust human pose refinement and a contact prediction module for efficient rendering.
- We demonstrate the effectiveness of HOGS on HOI rendering [24] and further extend it to hand-object grasp rendering [40], showcasing its applicability to diverse articulated interactions.

II. Related Work

Free Viewpoint Rendering. Free Viewpoint Rendering (FVR) focuses on synthesizing novel views of a scene and has been a long-standing challenge in computer vision [41], [42], [43], [44], [45]. Traditional methods generate novel views by blending or warping input views based on geometric constraints [46], [47], [48], but are limited to viewpoints near the input cameras. Recent research has shifted towards using neural representations, which enable higher-quality FVR both for static scenes and dynamic scenarios [49], [50], [51], [52]. However, these methods typically require dense views for higher-quality rendering. In real-world applications, dense-view setups are impractical or impossible, leading to sparse-view scenarios due to occlusions or limited camera setups [30], [53], [54], [55], [56]. These sparse-view conditions further complicate novel view synthesis, especially for complex HOIs. To address this challenge, our work focuses on high-quality HOI rendering from sparse views.

Differentiable Rendering of Radiance Fields. The advent of differentiable rendering has significantly advanced FVR [57], [58], [59]. Neural Radiance Fields (NeRF) [60] pioneered this field by representing scenes as implicit functions [51], [61], [62], [63], [64]. While NeRF provides impressive rendering results, its reliance on ray marching remains a computational bottleneck. In contrast, 3D Gaussian Splatting (3DGS) [25] employs explicit 3D Gaussians for fast and efficient rendering, rapidly emerging as a powerful technique for real-time rendering [5], [65], [66]. Leveraging its inherent efficiency, 3DGS has been successfully extended to dynamic human modeling, enabling the impressive reconstruction of human avatars [35], [36], [67]. However, these methods primarily focus on single humans and do not explicitly address the complex interactions inherent in multi-object scenes. In this work, we extend 3DGS to jointly model and render

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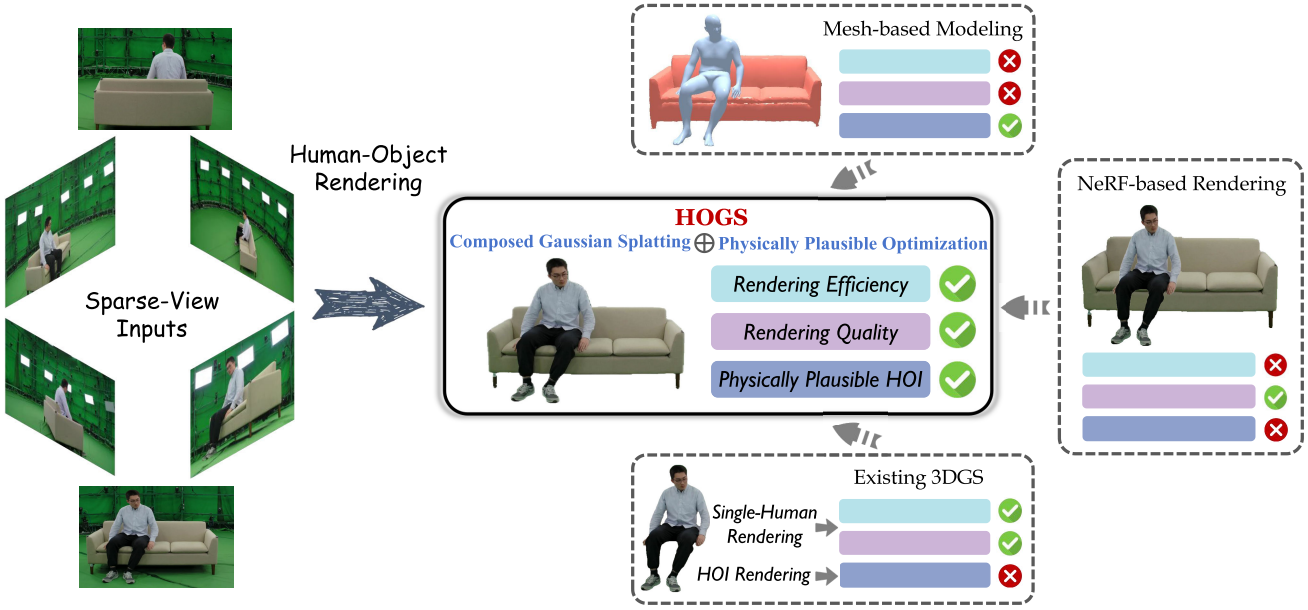


Fig. 1. **Comparison of state-of-the-art sparse-view HOI rendering methods.** Mesh-based methods have limitations in rendering efficiency and visual quality. NeRF-based methods, while capable of high fidelity, typically struggle with rendering efficiency and physically plausible HOI. Moreover, existing 3DGS methods lack effective HOI handling despite decent single-human rendering performance. In contrast, our proposed HOGS significantly improves high-quality, efficiency, and physically plausible HOI rendering simultaneously.

HOIs, enabling efficient and high-quality rendering of complex scenes.

HOI Rendering. Early approaches for HOI rendering primarily relied on mesh-based reconstructions, where humans and objects were reconstructed separately and then rendered together [18], [19], [20]. While straightforward, these methods often struggled with challenges such as occlusions and incomplete textures. More recent research has explored neural rendering techniques for HOIs, including texture blending [22], volumetric rendering [24], [68], and NeRF-based pipelines [14]. Among the latest advancements, NeuralDome [24] employs a layer-wise neural processing pipeline to render complex interactions. However, it relies heavily on dense multi-view inputs and computationally expensive volumetric rendering, making it less suitable for sparse-view scenarios.

Beyond visual fidelity, enforcing ***physically plausible*** interactions is crucial for realistic HOI rendering [69], [70]. Traditional methods often incorporated physical optimizations into their mesh-based reconstructions to ensure plausible human-object contact. This was typically achieved using penalty terms [71], [72], [73], specific constraints [74], [75], or physical criteria [76], [77] that operate on mesh vertices. However, these techniques, designed for static mesh topologies, are not directly applicable to modern explicit representations like 3D Gaussian Splatting. The 3DGS framework relies on a collection of Gaussian primitives that dynamically change in number and position during optimization, rendering mesh-centric constraints ineffective.

To bridge this gap, our work introduces HOGS, which not only leverages 3DGS for an efficient and high-fidelity solution to HOI rendering from sparse views but also pioneers a physically plausible optimization tailored for Gaussian primitives. By using novel Gaussian contact and separation losses, our method is the first to enforce physically plausible HOIs directly within the 3DGS framework, significantly enhancing both realism and rendering quality.

III. Methodology

For each frame of a dynamic HOI scene, our HOGS pipeline takes as input N sparse-view RGB images $\{I_i\}_{i=1}^N$, along with their corresponding human-object foreground masks $\{M_i^{ho}\}_{i=1}^N$. The pipeline operates on a hybrid optimization strategy (as shown in Figure 2): First, to overcome the ambiguity of sparse views, we leverage two powerful, generalizable modules (marked with ❄)—the *Sparse-View Human Pose Refinement* module and the *Sparse-View Human-Object Contact Prediction* module. These networks are pre-trained only once and function as robust off-the-shelf estimators to provide initial pose parameters and contact regions. Second, these estimations serve as the foundation for our per-scene optimization, where we employ a learnable *LBS Modulation* (marked with 🔥) and optimize the Gaussian attributes to recover fine-grained details. This strategy separates the costly training of generalizable models from the efficient, per-scene optimization of the final Gaussian representation. To provide a clear roadmap of this process, the step-by-step training procedure is formally summarized in Algorithm 1. The HOGS pipeline comprises three main stages, corresponding to the step-by-step data flow visualized in Figure 2 and detailed sequentially in Sec. III-A (Human-Object Deformation), Sec. III-B (Composed Gaussian Splatting), and Sec. III-C (Physically Plausible Optimization).

A. Human-Object Deformation

To accurately represent HOIs, we first deform both the human and the object from initial states to their respective target states. Human deformation includes applying standard LBS transformations, adding an LBS modulation to capture

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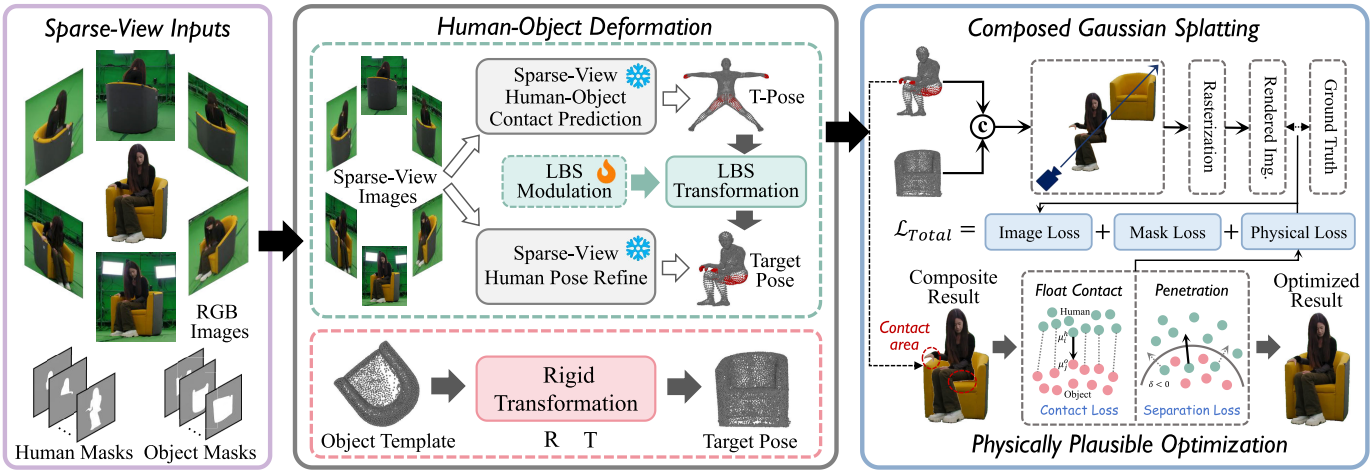


Fig. 2. **Overview of the HOGS pipeline.** Our framework takes sparse-view images of an HOI scene as input and consists of three main stages. (1) *Human-Object Deformation*: The canonical human and object models are transformed to match the current scene's pose. This process is driven by two key pre-trained modules processing the sparse inputs: the `Sparse-View Human Pose Refiner` (frozen ❄) aggregates sparse-view information to predict robust pose parameters, which then drive the `LBS Transformation`. Simultaneously, the learnable `LBS Modulation` (trainable 🔥) refines the skinning weights to capture non-rigid details. The object template is rigidly transformed to match the target pose. (2) *Composed Gaussian Splatting*: The posed models are converted into distinct sets of human and object Gaussians, which are then merged into a unified representation for joint, efficient rendering. (3) *Physically Plausible Optimization*: Finally, the rendering is refined to ensure physical plausibility. This stage utilizes the contact mask predicted by the `Sparse-View Human-Object Contact Prediction` module (frozen ❄) to focus the computation of our contact and separation losses on relevant interaction regions.

Algorithm 1 Overview of HOGS Pipeline

Input: Sparse images $\{I_i\}_{i=1}^N$, Masks $\{M_i^{ho}\}_{i=1}^N$, Pre-trained Modules (Refiner, Contact Predictor).
Output: Optimized Human-Object Gaussians $\mathcal{G}$.

```
// -- Initialization & Pre-computation --
Estimate human pose H_opt via Sparse-View Pose Refiner;
Estimate object pose (R_t, T_t) and obtain M_tar;
Compute SDF of M_tar for L_sep calculation;
Predict contact vertex set C via Contact Predictor;
Initialize human Gaussians G^h and object Gaussians G^o;
// -- Training Pipeline --
while not converged do
    // Human-Object Deformation
    Compute LBS modulation m = Φ_lbs(γ(p^c));
    Refine w_k and transform G^h to p^t via LBS (Eq. (1)–(3));
    // Composed Gaussian Splatting
    Compose unified scene: G ← {G^h} ∪ {G^o};
    Project G to 2D views and rasterize to rendered images Î;
    // Physically Plausible Optimization
    Compute rendering losses: L_image, L_ssim, L_lpips, L_mask;
    Compute L_contact on set C (Eq. 7);
    Compute L_sep using pre-computed SDF (Eq. 8);
    L_total ← Sum of weighted losses (Eq. 9);
    Update Gaussian parameters and LBS MLP Φ_lbs;
end
return Final G
```

finer details, and further refining the target pose through a refinement module. The object deformation is simplified by treating the object as rigid and estimating its rigid transformation with respect to a template mesh.

1) LBS Transformation: LBS is commonly employed in human rendering to deform human representations [34], [35], [36], [67]. Following this approach, we utilize LBS to transform human Gaussians. We adopt the SMPL-H [39] model, a parametric 3D human body model that extends SMPL to incorporate high-fidelity hand details. 3D Gaussians are initialized by placing their means $\mathbf{p}^c$ at the SMPL-H mesh vertices and assigning each an initial covariance $\mathbf{\Sigma}^c$. These initial Gaussians, defined in the canonical T-pose space, are then transformed to the posed space via LBS. Specifically, the transformed mean $\mathbf{p}^t$ and covariance $\mathbf{\Sigma}^t$ are given by:

$$\mathbf{p}^t = \sum_{k=1}^{K} w_k(\mathbf{R}_k\mathbf{p}^c + \mathbf{t}_k) + \mathbf{b}, \tag{1}$$

$$\mathbf{\Sigma}^t = \left(\sum_{k=1}^{K} w_k\mathbf{R}_k\right)\mathbf{\Sigma}^c\left(\sum_{k=1}^{K} w_k\mathbf{R}_k\right)^T, \tag{2}$$

where K is the number of joints, w_k is the LBS weight associated with joint k, $\mathbf{b}$ is a global translation vector, and $\mathbf{R}_k$ and $\mathbf{t}_k$ represent the rotation matrix and translation vector.

2) LBS Modulation: While LBS provides an efficient way to deform the human representation, it is a linear transformation strictly limited to the underlying mesh topology, Consequently, it often struggles to capture fine-grained details and subtle deformations. Inspired by previous works that learn LBS weight fields for detailed human modeling [23], [36], [78], [79], [80], we introduce an LBS modulation to refine the initial LBS weights derived from the SMPL-H model. Specifically, we leverage the pre-computed SMPL-H weights as a strong prior and employ an MLP Φ_{lbs} to modulate each LBS weight. Given a 3D Gaussian centered at $\mathbf{p}^c$ in the canonical space, we first apply a positional encoding $\gamma(\mathbf{p}^c)$ to its position. The MLP then outputs a modulation vector $\mathbf{m} = \Phi_{lbs}(\gamma(\mathbf{p}^c))$, where each element m_k corresponds to the modulation factor for the k-th LBS weight. The final LBS weight w_k is computed through a softmax function:

$$w_k = \texttt{softmax}(w_k^{\text{SMPL-H}} + m_k) \tag{3}$$

where $w_k^{\text{SMPL-H}}$ is the LBS weight derived from the nearest SMPL-H vertex for joint k. This approach modulates LBS weights efficiently, capturing finer details of deformations.

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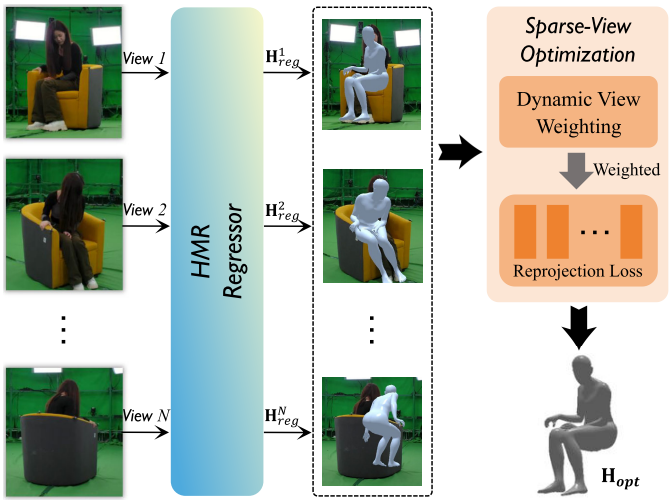


Fig. 3. **Training strategy for the Sparse-View Human Pose Refinement module.** We employ an optimization-in-the-loop approach to train our HMR-based regressor. For each training sample, the regressor's initial per-view estimates are refined by a differentiable multi-view optimization process, which leverages dynamic view weights to handle occlusions. The regressor is then trained end-to-end to directly predict this refined, multi-view consistent pose.

3) Sparse-View Human Pose Refinement: Accurate LBS transformations depend on a reliable target human pose, but achieving this from sparse views is challenging due to frequent human-object occlusions. Existing multi-view optimization methods typically rely on the triangulation of 2D keypoints or latent feature fusion [81], [82], [83], [84], [85], [86]. However, triangulation degrades severely when 2D detections are noisy due to occlusion, while feature fusion struggles to generalize to sparse camera setups without extensive pre-training. To overcome these limitations, we develop a Sparse-View Human Pose Refinement module. Unlike methods that implicitly learn fusion weights, we introduce a ***Dynamic View Weighting*** mechanism to explicitly filter out occluded noise based on physical occlusion rates. This ensures that the optimization is strictly driven by reliable visual cues. Furthermore, rather than performing costly test-time optimization for each new scene, our goal is to build a powerful, generalizable regressor that can directly infer high-quality poses in a single forward pass. As illustrated in Figure 3, this strategy leverages a multi-view optimization process during the training phase to teach a standard regressor how to handle occlusions and fuse information from sparse views.

Training Strategy. Our training strategy revolves around an HMR-based regressor [87] and a novel, optimization-based training objective. For each training sample, the regressor first produces an initial set of SMPL-H parameters $\mathrm{H}^i_{\mathrm{reg}} = \{\theta^i_{\mathrm{reg}}, \beta^i_{\mathrm{reg}}\}$ for each of the N sparse views. Instead of using these predictions directly in a simple loss, we use them to initialize a differentiable optimization process that finds a more physically plausible, multi-view consistent pose, which we denote as $\mathrm{H}_{\mathrm{opt}}$. The final loss function then encourages the regressor's direct output, $\mathrm{H}^i_{\mathrm{reg}}$, to be as close as possible to the target pose $\mathrm{H}_{\mathrm{opt}}$.

The core of this strategy lies in defining the optimization objective for finding $\mathrm{H}_{\mathrm{opt}}$. This process is inspired by SMPLify [88] and is adapted for our sparse-view setting by aggregating per-view costs. The cost for the i-th view is:

$$\mathcal{E}_i = \|P(\mathrm{H}) - J^i_{\mathrm{reg}}\|^2 + \lambda_\theta E_\theta(\theta) + \lambda_\beta E_\beta(\beta), \tag{4}$$

where $P(\mathrm{H})$ is the 2D projection of the estimated 3D joints, J^i_{reg} are the initial 2D joints predicted by the regressor, and $E_\theta(\theta), E_\beta(\beta)$ are regularization terms [89] with weights $\lambda_\theta = 1$ and $\lambda_\beta = 0.001$.

Dynamic View Weighting. To handle the varying reliability of different views due to occlusions, we introduce dynamic view weights d_i. We estimate an occlusion rate $O_i = 1 - V_i/K$, where V_i is the number of visible joints within the human mask in view i and K is the total joint count. These rates are used to compute the weights via $d_i = \mathtt{softmax}(-\alpha O_i)$, where $\alpha = 5$ is a sensitivity factor. The total optimization cost is then the weighted sum $\mathcal{E}_{\mathrm{total}} = \sum_{i=1}^N d_i \cdot \mathcal{E}_i$. By minimizing this cost, we obtain the optimized target pose $\mathrm{H}_{\mathrm{opt}} = \{\theta_{\mathrm{opt}}, \beta_{\mathrm{opt}}\}$. It is crucial to note that this optimization is a differentiable operation ***within the training loop***, serving to produce a high-quality supervision signal for our regressor.

End-to-End Regressor Training. The HMR-based regressor is trained end-to-end to minimize a composite loss function that leverages the optimized pose target $\mathrm{H}_{\mathrm{opt}}$:

$$\mathcal{L}_{\mathrm{HPR}} = \lambda_1 \mathcal{L}_{2D} + \lambda_2 \mathcal{L}_{3D} + \lambda_3 \mathcal{L}_{\mathrm{H}}, \tag{5}$$

where $\mathcal{L}_{2D} = \sum_{i=1}^N \|J^i_{\mathrm{reg}} - J^i_{\mathrm{gt}}\|$ is a standard 2D reprojection loss. The 3D loss $\mathcal{L}_{3D} = \sum_{i=1}^N \|\mathbf{p}^i_{\mathrm{opt}} - \mathbf{p}^i_{\mathrm{gt}}\|$ supervises the optimized 3D joints $\mathbf{p}^i_{\mathrm{opt}}$ against the ground truth. Most importantly, the consistency loss $\mathcal{L}_{\mathrm{H}} = \sum_{i=1}^N \|\mathrm{H}^i_{\mathrm{reg}} - \mathrm{H}_{\mathrm{opt}}\|$ explicitly forces the regressor's direct output to match the pose found by the optimization process. In this way, the network learns to internalize the logic of multi-view fusion and occlusion handling. The loss weights are set to $\lambda_1 = 5$, $\lambda_2 = 5$, and $\lambda_3 = 0.001$.

Inference on New Scenes. The described optimization-guided training strategy yields a pose refinement module that is both powerful and efficient. At inference time, for any new or unseen scene, we perform a ***single forward pass*** through the trained regressor to obtain the refined human pose parameters. Crucially, no per-scene optimization is required, which enables robust and efficient pose estimation.

4) Object Deformation: Objects involved in HOIs are typically rigid. Following prior work [24], [90], we estimate object rotation $R_t \in SO(3)$ and translation $T_t \in \mathbb{R}^3$ relative to a template mesh $\mathcal{M}_{\mathrm{temp}}$. As shown in Fig. 2, we employ ICP [91] to register $\mathcal{M}_{\mathrm{temp}}$ to per-frame 3D object markers. These markers ensure robust ICP initialization, mitigating local minima issues and yielding accurate per-frame poses (R_t, T_t) for the target object mesh $\mathcal{M}_{\mathrm{tar}}$. This enables tracking the rigid motion of objects. It is worth noting that we strategically employ this stable and established rigid solver to ensure system stability. This design choice allows us to allocate computational resources to our core innovation—the Physically Plausible Optimization—which resolves the more critical and challenging issue of human-object interaction.

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B. Composed Gaussian Splatting

We extend 3DGS to jointly render humans and objects. While treating the scene as a single Gaussian set might seem straightforward, it leads to ***structural conflicts*** due to the disparate characteristics of the entities: the high-frequency motion gradients from the articulated human often trigger erroneous densification in the adjacent rigid object. To resolve this, our strategy adheres to two design principles: (1) ***Decoupled Learning Dynamics***, where human and object Gaussians are maintained as distinct subsets to preserve their specific structural properties (i.e., articulated vs. rigid); and (2) ***Unified Differentiable Sorting***, where these subsets are merged prior to rasterization. This ensures that despite being maintained separately, they enter the same visibility sorting process, which is crucial for resolving depth ambiguities and occlusion relationships in sparse-view settings.

Based on this design, we model the scene with two distinct Gaussian subsets: 1) ***Human Gaussians***: For the human, each vertex of the SMPL-H model is converted into a 3D Gaussian, forming the human Gaussian set $\{\mathcal{G}_i^h = \mathcal{N}(\mathbf{x}_i; \mu_i^h, \Sigma_i^h)\}$, where $\mathcal{N}(\mathbf{x}; \mu, \Sigma)$ represents a Gaussian distribution with mean μ and covariance Σ. The human Gaussian parameters μ_i^h and Σ_i^h are updated frame-by-frame based on the refined pose and shape parameters from Sec. III-A, ensuring accurate motion tracking of the articulated human body. 2) ***Object Gaussians***: For the object, the vertices of the target pose of object mesh $\mathcal{M}_{\text{tar}}$ are used to initialize the means of the object Gaussians. Specifically, each vertex of $\mathcal{M}_{\text{tar}}$ becomes the mean μ_j^o of an object Gaussian $\mathcal{G}_j^o$, with an assigned initial covariance Σ_j^o. Finally, the human and object Gaussian sets are merged into a unified set: $\mathcal{G} = \{\mathcal{G}_i^h\} \cup \{\mathcal{G}_j^o\}$.

The combined Gaussian set $\mathcal{G}$ is projected into the 2D plane. The covariance matrix Σ of each 3D Gaussian is transformed to its projected 2D covariance Σ' as follows:

$$\Sigma' = \mathbf{J}\,\mathbf{W}\,\Sigma\,\mathbf{W}^T\mathbf{J}^T, \tag{6}$$

where $\mathbf{W}$ is the world-to-camera transformation matrix, $\mathbf{J}$ is the Jacobian of the affine approximation of the projective transformation. The projected Gaussians are blended using α-compositing to generate the rendered image, which is compared with the ground-truth image using $\mathcal{L}_{\text{image}}$ loss.

C. Physically Plausible Optimization

To ensure physically plausible HOI rendering, we introduce a two-step rendering optimization process. Firstly, we predict potential human-object contact regions using a novel sparse-view human-object contact prediction module. This prediction yields a set of human Gaussians corresponding to the contact regions. Secondly, leveraging these predicted regions, we perform a physical optimization on the composed Gaussian splatting result to enforce physical constraints.

1) Sparse-View Human-Object Contact Prediction: Directly compositing Gaussians (as in Sec. III-B) may lead to physically implausible interactions at human-object contact regions. However, enforcing physical constraints globally is computationally prohibitive, as it would necessitate calculating signed distance fields (SDF) for all dynamic Gaussians at every iteration. To ensure optimization feasibility, our strategy is to first efficiently identify the specific human-object contact regions. By doing so, we focus the subsequent physical optimization only on the relevant subset of primitives, avoiding expensive global queries. To this end, we propose a sparse-view contact prediction module, which is illustrated in Fig. 4.

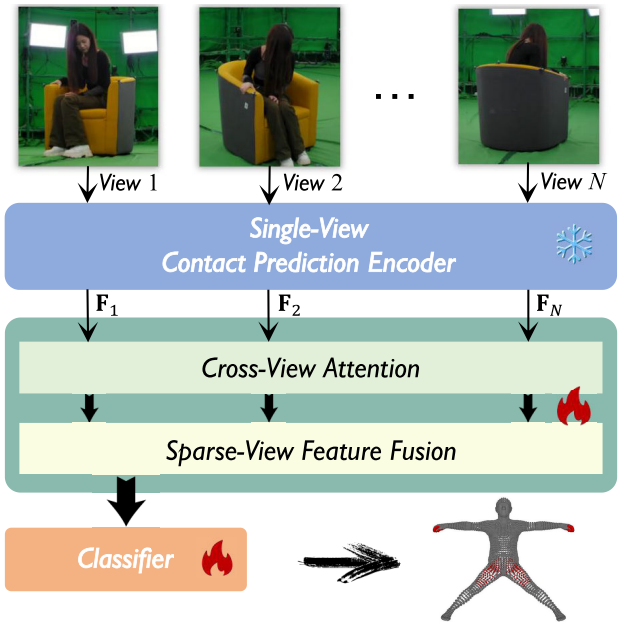


Fig. 4. **Architecture of the Sparse-View Contact Prediction module.** Given sparse-view images, features are first extracted from each sparse input view using a shared encoder. A ***cross-view attention*** module then fuses these features to aggregate multi-view information. Finally, a classifier predicts a per-vertex contact probability map on the human mesh to identify regions of interaction.

Architecture. Our module first extracts per-view HOI features from the N input views using a single-view contact prediction encoder. Specifically, we adapt the DECO [92] architecture, excluding its final MLP layer, to produce per-view features $\mathrm{F} = \{\mathrm{F}_i\}_{i=1}^N \in \mathbb{R}^{N\times D}$, where the feature dimension $D = 2048$. These features F are then processed by a cross-view attention module to aggregate information across all views. Within the attention module, the query, key, and value matrices ($\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$) are computed by linearly projecting the input features F using weight matrices $W_Q, W_K, W_V \in \mathbb{R}^{2048\times 512}$. The attention mechanism outputs a tensor $\mathrm{F}_{\text{att}} \in \mathbb{R}^{N\times 512}$. Subsequently, a sparse-view feature fusion module averages F_{att} across the view dimension to produce a single fused feature vector $\mathrm{F}_{\text{fuse}} \in \mathbb{R}^{512}$. Finally, a lightweight classifier takes F_{fuse} as input to yield contact probabilities $\mathcal{P} \in \mathbb{R}^{6890}$ for all body vertices of the SMPL-H model.

Training Process. The module is trained in a supervised manner. We adopt a transfer learning approach by initializing the single-view encoder with weights from the pre-trained DECO model [92]. During our training phase, the parameters of this encoder are frozen to retain its powerful feature extraction capabilities. We then train only the newly added cross-view attention module, the feature fusion module, and the final classifier. The training objective is a standard binary

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cross-entropy loss between the predicted probabilities $\mathcal{P}$ and the ground truth labels.

Inference Stage. At inference time, the pre-trained contact prediction module takes the N sparse views of a new scene as input. It performs a single forward pass to directly output the contact probabilities $\mathcal{P}$. These probabilities are then thresholded (with $\tau = 0.5$) to identify the contact vertex set $\mathcal{C}$, which is subsequently used to guide the physically plausible optimization (Sec. III-C.2).

2) Physically Plausible Optimization: In this context, we focus on establishing geometric consistency by specifically enforcing constraints such as non-penetration and surface contact. To enforce these constraints, previous mesh-based methods [16], [93], [94], [95] typically rely on vertex-to-vertex distances calculated via Nearest Neighbor (NN) search. However, directly applying such mesh-centric losses to 3DGS is computationally prohibitive. Unlike meshes with fixed semantic vertices (~6K), 3DGS involves massive, unstructured primitives (~10k-100K) that change dynamically. An explicit NN search for every Gaussian would result in intractable complexity ($O(N \log M)$). To address this, we propose a scalable alternative: we replace explicit neighbor searching with efficient Signed Distance Field (SDF) queries ($O(1)$ per Gaussian) and utilize our predicted contact mask to filter optimization targets. This combination allows us to enforce physical plausibility with negligible computational overhead.

With human Gaussians in contact regions identified (yielding the vertex index set $\mathcal{C}$ from contact prediction, Sec. III-C.1), we perform physically plausible optimization on these regions within our HOGS framework (Figure 2). This optimization enforces physical plausibility by minimizing our *Gaussian Contact and Separation Losses*, guiding human and object Gaussians towards plausible configurations.

Contact Loss. To encourage closer human-object Gaussian proximity in regions $\mathcal{C}$, we define the contact loss. Specifically, this loss considers distances between human Gaussians $\{\mathcal{G}_i^h(\mu_i^h, \Sigma_i^h)\}_{i\in\mathcal{C}}$ (for contact vertices in $\mathcal{C}$) and all N_o object Gaussians $\{\mathcal{G}_j^o(\mu_j^o, \Sigma_j^o)\}_{j=1}^{N_o}$, which is defined as:

$$\mathcal{L}_{\text{contact}} = \frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}} \min_{1\le j\le N_o} \|\mu_i^h - \mu_j^o\|_2 + \frac{1}{N_o}\sum_{j=1}^{N_o} \min_{i\in\mathcal{C}} \|\mu_j^o - \mu_i^h\|_2. \quad (7)$$

This loss term effectively pulls the interacting regions of the human and object Gaussians closer, promoting a more physically plausible interaction by reducing unnatural gaps.

Separation Loss. Besides reducing unnatural gaps via the contact loss, intuitive physics dictates that objects cannot occupy the same 3D space, a phenomenon known as *penetration*. Thus, we introduce a separation loss $\mathcal{L}_{\text{sep}}$ to penalize human-object Gaussian interpenetration. Our approach leverages Signed Distance Fields (SDFs) for efficient penetration detection. Directly computing SDFs from all dynamic Gaussians incurs a significant computational burden, as SDFs would need frequent recomputation during optimization due to changes in Gaussian numbers and positions. To mitigate this, we exploit the static nature of the object, represented by a fixed target mesh $\mathcal{M}_{\text{tar}}$ (Sec. III-A). Consequently, we pre-compute the SDF of $\mathcal{M}_{\text{tar}}$ only once before 3DGS optimization.

We construct a uniform $256 \times 256 \times 256$ voxel grid over a padded bounding box of $\mathcal{M}_{\text{tar}}$. Each voxel center $c_t \in \mathbb{R}^3$ stores its signed distance δ_t to the nearest surface point on $\mathcal{M}_{\text{tar}}$, where the sign of δ_t indicates if c_t is inside (negative) or outside (positive) the object. Penetration of a human Gaussian $\mathcal{G}_i^h$ is determined by querying the SDF at its mean μ_i^h. Trilinear interpolation within the grid yields a continuous signed distance $\delta(\mu_i^h)$ and normal $\mathbf{n}(\mu_i^h)$ at μ_i^h. The separation loss is then defined as:

$$\mathcal{L}_{\text{sep}} = \frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}} \max(0, -\delta(\mu_i^h))\|\mathbf{n}(\mu_i^h)\|_2^2, \quad (8)$$

where we only penalize negative $\delta(\mu_i^h)$ values (points inside the object).

Total Loss. As shown in Figure 2, the total training loss for composed Gaussian splatting is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{image}} + \lambda_{\text{ssim}}\mathcal{L}_{\text{ssim}} + \lambda_{\text{lpips}}\mathcal{L}_{\text{lpips}} + \lambda_{\text{mask}}\mathcal{L}_{\text{mask}} + \lambda_{\text{contact}}\mathcal{L}_{\text{contact}} + \lambda_{\text{sep}}\mathcal{L}_{\text{sep}}, \quad (9)$$

where $\mathcal{L}_{\text{mask}}$ uses human-object foreground masks $\{M_i^{ho}\}_{i=1}^N$, while $\mathcal{L}_{\text{image}}$ (Sec. III-B), SSIM loss [96], and LPIPS loss [97] ensure image fidelity against ground truth.

IV. Experiments

A. Experimental Settings and Implementation Details

1) Datasets: We conduct experiments on two datasets: the large-scale HOI rendering dataset **HODome** [24] and the hand-object grasp rendering dataset **MANUS-Grasps** [40]. **HODome** features 76 dynamic HOI sequences with 23 objects and 10 subjects. The dataset provides multi-view RGB videos, human/object masks, object templates, and initial SMPL-H parameters. For our experiments, we follow a sparse-view protocol, using 6 fixed input views to reconstruct each interaction scene. **MANUS-Grasps** is used to demonstrate HOGS's applicability to more fine-grained articulated interactions. It contains high-fidelity hand-object grasp sequences captured by 53 cameras. Since this dataset does not provide pre-scanned object templates, we showcase the flexibility of our method by directly using its well-trained object Gaussians, bypassing our object deformation module. Similarly, for MANUS-Grasps, we adopt the same sparse-view protocol following [24], utilizing 6 fixed input views for reconstruction.

2) Evaluation Metrics: To quantitatively evaluate the quality of rendered novel view and novel pose images, we evaluate the rendering quality using standard metrics following NeuralDome [24]: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [96], and Learned Perceptual Image Patch Similarity (LPIPS) [97].

3) Pre-Training of Generalizable Modules: As detailed in Sec. III, our pose refinement and contact prediction modules are pre-trained to be generalizable. We pre-train both modules on the HODome dataset. To rigorously evaluate generalization to unseen individuals, we partition the 9 available subjects into distinct training, validation, and testing sets. Specifically, our training set comprises 6 subjects (`01, 02, 03, 04, 06,`

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07). The validation set contains subject 08, used for hyper-parameter tuning and model selection. The test set consists of 2 completely unseen subjects (09, 10) for performance reporting. For the *Sparse-View Human Pose Refinement* module, we use the Adam optimizer with a learning rate of 3×10^{-5} and train for 20 epochs. For the *Sparse-View Human-Object Contact Prediction* module, we use the Adam optimizer with a learning rate of 1×10^{-5} and train for 100 epochs.

4) HOGS Implementation Details: The following details pertain to the per-scene optimization of the main HOGS pipeline. All experiments are conducted on a single NVIDIA A100 GPU. The *LBS Modulation* module (Sec. III-A) is an MLP with three hidden layers and ReLU activations. It takes a 63-dimensional positional encoding of the canonical Gaussian means as input and outputs a 52-dimensional modulation vector. This MLP is optimized with the rest of the framework using the Adam optimizer with a learning rate of 1×10^{-5}. For the *Composed Gaussian Splatting* stage, other 3DGS-related training details (e.g., initialization, densification, pruning, optimization schedule) follow the original implementation [25]. The weights for the total loss function (Sec. III-C.2) are set as follows: $\lambda_{\text{mask}} = 0.3$, $\lambda_{\text{ssim}} = 0.5$, $\lambda_{\text{lpips}} = 0.1$, $\lambda_{\text{contact}} = 0.01$, and $\lambda_{\text{sep}} = 0.01$. These hyperparameters, particularly the physical loss weights, are determined through a sensitivity analysis on the validation set (Subject 08) to achieve the optimal balance between geometric plausibility and visual fidelity (see Sec. IV-C for detailed analysis).

TABLE I

QUANTITATIVE RESULTS OF NOVEL VIEW SYNTHESIS ON HODOME DATASET. NH-FVV DENOTES NEURALHUMANFVV, AND NHOI-FVV DENOTES NEURALHOIFVV. WE USE RED AND YELLOW TEXT TO DENOTE THE BEST AND SECOND-BEST RESULTS OF EACH METRIC RESPECTIVELY

Category	Method	PSNR ↑	SSIM ↑	LPIPS ↓	FPS ↑
Mesh-based	NH-FVV	21.69	0.914	-	12
NeRF-based	IBRNet	21.43	0.892	-	-
	NeuRay	23.34	0.909	-	2
	NHOI-FVV	23.10	0.912	-	1
3DGS-based	MANUS	27.28	0.936	0.062	155
	HOGS (Ours)	30.68	0.953	0.028	162

TABLE II

QUANTITATIVE RESULTS ON THE MANUS-GRASPS DATASET. WE EVALUATE HOGS AGAINST RECENT STATE-OF-THE-ART METHODS MANUS, HOLD, AND BIGS. WE USE RED AND YELLOW TO DENOTE THE BEST AND SECOND-BEST RESULTS

Method	PSNR ↑	SSIM ↑	LPIPS ↓
HOLD [101]	25.913	0.9866	0.0722
MANUS [40]	26.328	0.9872	0.0688
BIGS [102]	26.948	0.9885	0.0690
HOGS (Ours)	27.425	0.9891	0.0651

B. Comparisons With State-of-The-Arts

1) Comparison Methods: We compare HOGS against a comprehensive set of state-of-the-art methods to evaluate its performance on both human-object and hand-object rendering.

For human-object rendering on HODome, comparison methods cover three main paradigms. **Mesh-based:** We include NeuralHumanFVV (NH-FVV) [17], a representative method that learns to blend textured human reconstructions. **NeRF-based:** We compare against a suite of strong NeRF models, including the general novel-view synthesizer IBRNet [98]; NeuRay [99], which explicitly models visibility to handle occlusions; and NeuralHOIFVV (NHOI-FVV) [24], a state-of-the-art HOI-specific method using a layered NeRF representation. **3DGS-based:** To create a challenging 3DGS competitor, we adapt **MANUS** [40]. As MANUS is originally designed for hands, we adapt it for full-body HOIs by replacing its hand component with our human deformation module while retaining its core object rendering pipeline.

For hand-object rendering on MANUS-Grasps, we compare HOGS against its original baseline, MANUS [40], as well as two recent leading methods in hand-object rendering: HOLD [100] and BIGS [101]. This allows for a thorough evaluation of HOGS's extensibility and performance on fine-grained interactions.

2) Results on Human-Object Rendering: **Quantitative Analysis.** Table I presents the quantitative comparison on the HODome dataset. Our proposed HOGS significantly outperforms all baselines across all evaluation metrics. Specifically, HOGS achieves a PSNR of 30.68 dB, surpassing the next-best 3DGS-based method by a large margin of 3.4 dB. This substantial improvement in rendering quality highlights the effectiveness of our composed Gaussian representation and physically plausible optimization. In terms of efficiency, HOGS achieves a rendering speed of 162 FPS, which is not only the fastest among all methods but also orders of magnitude faster than NeRF-based approaches (1)-2 FPS), demonstrating its suitability for real-time applications.

Qualitative Analysis. Figure 5 presents qualitative comparisons of HOGS with existing methods. HOGS achieves superior rendering quality, generating novel views that closely match reference images. In contrast, the mesh-based NH-FVV struggles with occlusions, leading to noticeable artifacts and incomplete geometry. NeRF-based methods also exhibit various artifacts like blurry textures and geometric distortions. While the 3DGS-based MANUS captures overall human-object structure with reasonable rendering results, it lacks fine details in HOI regions.

3) Extensibility to Hand-Object Grasping: To demonstrate the flexibility and scalability of our framework, we extend HOGS to the hand-object grasping task and evaluate it on the MANUS-Grasps dataset. To adapt our method for this task, we substitute the SMPL-H full-body model in our human deformation stage (Sec. III-A) with the widely-used MANO hand model [102], while keeping all other components of our pipeline unchanged. This simple adaptation highlights the modularity of our design.

Quantitative Analysis. As shown in Table II, HOGS consistently achieves the best performance even when compared against state-of-the-art methods specifically designed for hand-object capture. Our method surpasses the recent strong

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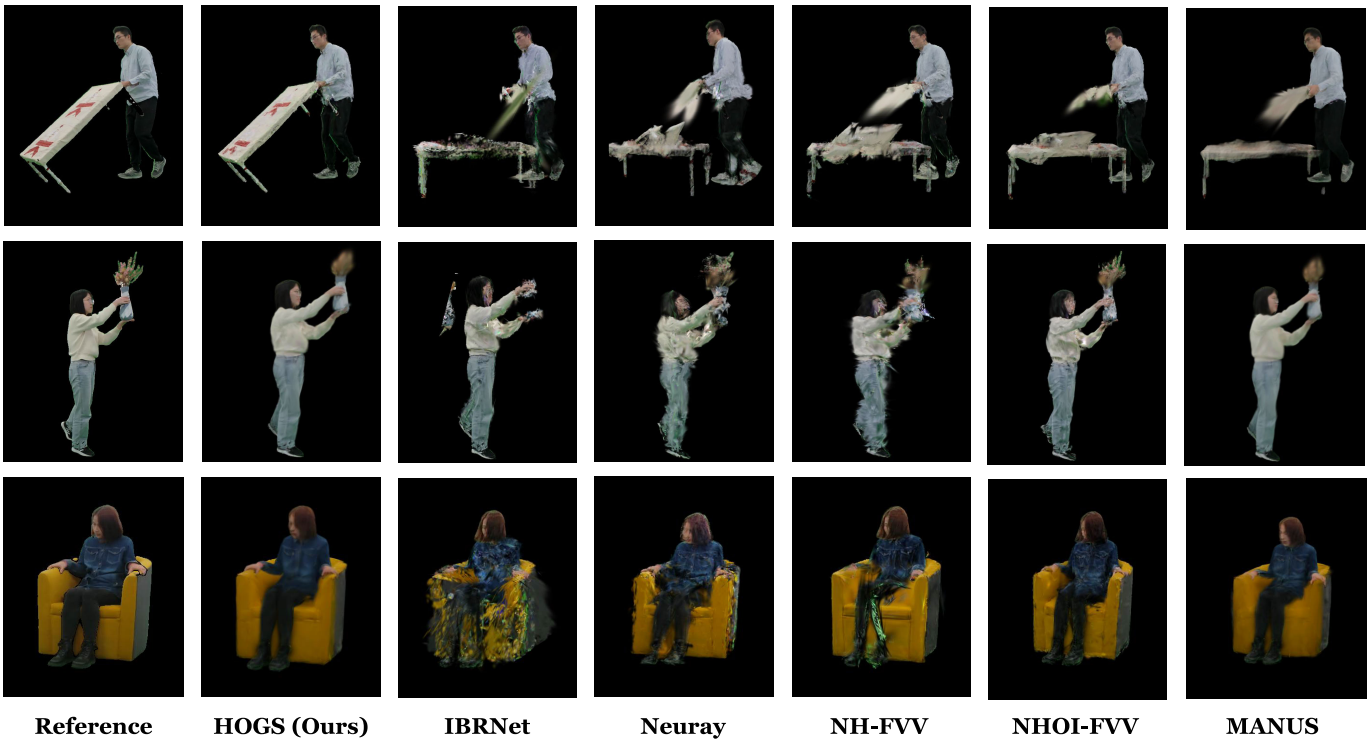


Fig. 5. **Qualitative comparison on the HODome dataset.** HOGS renders novel views with superior visual fidelity and more plausible HOIs compared to existing methods. In contrast, NeRF-based approaches (*e.g.*, IBRNet, NeuRay) suffer from significant artifacts like blurriness and geometric distortions. The mesh-based NH-FVV exhibits incomplete geometry, while the adapted 3DGS-based MANUS, although structurally sound, lacks fine-grained detail, especially in contact regions.



Fig. 6. **Extensibility of HOGS.** Rendering results of diverse hand-object grasping scenarios from the MANUS-Grasps dataset.

baselines across all metrics, achieving a PSNR of 27.425 and an LPIPS score of 0.0651. This validates the effectiveness and strong generalization capability of our core pipeline for rendering complex, articulated interactions.

Qualitative Analysis. The visual results in Figure 6 further showcase HOGS's capability. Our method successfully renders a variety of challenging hand-object grasping scenes, capturing subtle details of finger-object contact and realistically representing diverse object shapes. These results confirm that our HOGS framework is not limited to full-body interactions but is broadly applicable to diverse articulated scenarios, demonstrating strong generalization capabilities across different scales of interaction.

C. Component Analysis

In this section, we conduct a series of analytical experiments to validate the effectiveness of each key component within our HOGS framework.

1) Ablation Study of Core Components: We perform a comprehensive ablation study to demonstrate the cumulative contribution of our proposed modules. Starting from a basic setup, we incrementally add each component and report the performance on the HODome dataset (subject01 subset). The results are summarized in Table III.

TABLE III

ABLATION STUDY OF HOGS COMPONENTS. WE START WITH A BASELINE AND INCREMENTALLY ADD OUR PROPOSED MODULES: LBS MODULATION, SPARSE-VIEW (SV) HUMAN POSE REFINEMENT, COMPOSED GAUSSIAN SPLATTING, PHYSICAL LOSS, AND SV CONTACT PREDICTION. WE USE RED AND YELLOW TEXT TO DENOTE THE BEST AND SECOND-BEST RESULTS OF EACH METRIC RESPECTIVELY

Method	PSNR ↑	SSIM ↑	FPS ↑
Baseline	26.96	0.928	179
+ LBS Modulation	26.98	0.928	175
+ SV Human Pose Refinement	27.40	0.931	174
+ Composed Gaussian Splatting	29.12	0.944	171
+ Contact Loss	29.95	0.951	165
+ Separation Loss	30.85	0.956	157
+ SV Contact Prediction	30.47	0.949	162

Baseline Setup. Our baseline renders human and object via separate 3D Gaussian Splatting, then combines their point clouds for final rendering. In this baseline, human deformation solely relies on the LBS transformation (Sec. III-A) with a randomly selected pose from the 6 input views as the target pose, while object deformation is performed consistently with the method detailed in Sec. III-A.

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(i) w/o LBS Modulation (ii) w LBS Modulation

Fig. 7. **Visual ablation of LBS Modulation.** (i) Standard LBS results in over-smoothed, blurry boundaries. (ii) Our LBS Modulation recovers sharp details and corrects clothing geometry by breaking the linearity of SMPL.

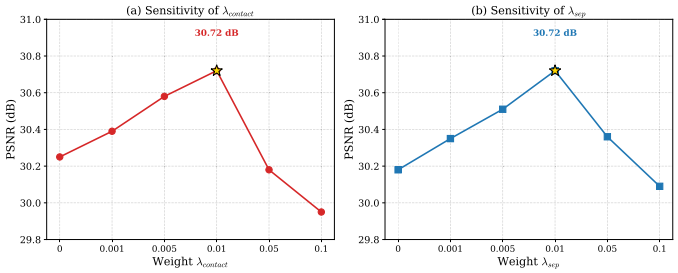


Fig. 8. **Sensitivity analysis of physical loss weights on the validation set.** Both (a) Contact Loss and (b) Separation Loss exhibit a clear inverted-U shaped trend, confirming $\lambda = 0.01$ as the optimal balance point between geometric constraints and visual fidelity.

TABLE IV

COMPARISONS OF DIFFERENT HUMAN-OBJECT RENDERING STRATEGIES ON THE HODOME DATASET

Strategy	PSNR ↑	SSIM ↑
Unified Optimization (Single Set)	28.90	0.941
Separate Rendering & 2D Compositing	28.47	0.935
Ours (Composed Gaussian Splatting)	**29.12**	**0.944**



(i) Unified Optimization (ii) Separate Rendering (iii) Ours

Fig. 9. **Visual comparison of different rendering strategies.** (i) Unified Optimization suffers from severe artifacts and blurring on the rigid sofa. (ii) Separate Rendering leads to unnatural depth boundaries and edge artifacts. (iii) Ours preserves the sharp geometry of the rigid object while correctly handling the human-object depths.

Incremental Module Analysis. We incrementally add our modules to the baseline, evaluating novel view synthesis. As shown in Table III, rendering quality consistently improves with each added module, peaking with the Separation Loss. This progressive improvement demonstrates the effectiveness of our designed modules.

Impact of LBS Modulation. Regarding the LBS Modulation, although the quantitative gain in Table III is subtle, its visual impact is critical. While standard LBS is computationally efficient, as a linear transformation, it is strictly limited to the underlying mesh topology, which often struggles to capture fine-grained details and subtle deformations essential for realistic rendering. Our LBS Modulation acts as a learnable residual field designed to break this linearity. As highlighted in Figure 7, standard LBS results in over-smoothed, blurry boundaries (e.g., around the legs). In contrast, our modulation successfully recovers sharp details and corrects clothing geometry.

Impact of Sparse-View Contact Prediction Module. Incorporating sparse-view contact prediction yields a slight quality decrease but significantly boosts rendering efficiency (Table III). This stems from focused physical loss computation. Without contact prediction, our *Gaussian Contact and Separation Losses* (Eq. (7) and Eq. (8)) apply to all human-object Gaussian pairs—a thorough but computationally expensive process. However, HOI contact typically occurs in limited regions, involving partial Gaussians. Thus, sparse-view contact prediction is crucial for efficient HOI rendering, letting HOGS focus optimization on relevant contact regions to improve computational efficiency.

Sensitivity of Physical Loss Weights. To validate our hyperparameter choice for physical constraints, we perform a sensitivity analysis on the validation set (Subject 08), as shown in Figure 8. We employ a control variate method, varying one weight (λ_{contact} or λ_{sep}) while fixing others. Both losses exhibit a clear inverted-U shaped trend, peaking at 0.01. Specifically, increasing λ_{contact} up to 0.01 steadily improves PSNR by eliminating floating artifacts, but excessive weights (>0.05) degrade performance by unnaturally compressing geometry. Similarly, $\lambda_{\text{sep}} = 0.01$ provides the optimal balance for resolving inter-penetration without causing Gaussians to vanish (as observed with larger weights like 0.1). This confirms that our chosen weights achieve the best trade-off between geometric plausibility and visual fidelity.

2) Analysis of Rendering Strategies: A core design choice in HOGS is *Composed Gaussian Splatting*, which maintains separate Gaussian sets for human and object but renders them jointly. To verify the necessity of this strategy, we compare it against two alternatives: (1) ***Unified Optimization***, where human and object are treated as a single homogeneous Gaussian set; and (2) ***Separate Rendering***, where entities are rendered into separate images and composited using depth maps.

As shown in Table IV, our strategy outperforms both baselines. Unified optimization suffers from *dynamic conflict*—high gradients from the moving human cause artifacts on the rigid object (visualized in Figure 9). Separate rendering fails to resolve complex occlusion boundaries in 3D space, leading to unnatural edges (-0.65 dB). Our approach effectively decouples the learning dynamics while ensuring correct visibility sorting, proving it is a vital component for high-fidelity HOI rendering.

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Fig. 10. **Effectiveness of the SV Human Pose Refinement Module.** (a) Without refinement, using a pose from a single, potentially occluded view leads to incorrect body posture. (b) Our refined pose corrects these errors, resulting in a physically plausible and accurate human reconstruction.

TABLE V
COMPARISON OF MULTI-VIEW HUMAN POSE ESTIMATION ACCURACY (MPJPE/PA-MPJPE IN MM) ON THE HODOME DATASET

Method	MPJPE ↓	PA-MPJPE ↓
Qiu et al. [82]	83.5	69.7
AdaFuse [87]	71.0	61.6
MvP [104]	71.3	61.4
Ours (Learnable α)	69.6	60.8
Ours (Fixed $\alpha = 5$)	**69.4**	**60.8**

3) Effectiveness of Pre-Trained Modules: We provide qualitative results to further validate the two core pre-trained modules that enable robust performance from sparse views.

Figure 10 illustrates the impact of our ***human pose refinement*** module. As shown, relying on an off-the-shelf pose estimator [87] from a single view can result in severe geometric artifacts due to occlusions (Fig. 10(a)). Our module effectively fuses information from all sparse views to produce a coherent and accurate 3D pose, leading to a much more realistic human rendering (Fig. 10(b)). To further quantitatively validate our design, we compared our module against state-of-the-art multi-view pose estimation methods [81], [86], [103] on the HODome dataset. As reported in Table V, our method achieves the lowest error with an MPJPE of 69.4 mm, outperforming the best baseline by a clear margin. Unlike standard fusion methods that treat views equally, our dynamic view weighting effectively filters out occluded noise, which is critical for robust HOI initialization. Moreover, we evaluate a variant where the sensitivity factor α is set as a learnable parameter. The learnable α yields a higher MPJPE than our fixed setting. This suggests that a fixed α effectively enforces the physical prior that occluded views are less reliable, preventing potential overfitting to noise in sparse-view inputs.

Figure 11 showcases the effectiveness of our ***human-object contact prediction*** module. It can precisely locate the contact regions between the human and the object, from large contact surfaces (e.g., sitting on a sofa) to fine-grained interactions (e.g., hand touching an object). This accurate prediction serves as a critical prerequisite for ensuring the efficiency of our physically plausible optimization.

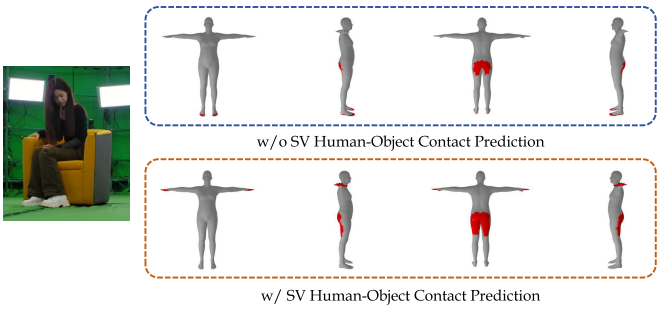


Fig. 11. **Effectiveness of the SV Human-Object Contact Prediction Module.** Our module accurately identifies contact regions (highlighted in red) from sparse views, even for challenging interactions like sitting (left) and fine-grained hand contact (right), enabling focused and efficient physics optimization.

TABLE VI
QUANTITATIVE EFFECT OF OUR PHYSICAL LOSSES, MEASURED BY THE MEAN SQUARED PENETRATION DEPTH (MSPD)

Method	MSPD (cm^2) ↓
HOGS w/o Physical Losses	3.829
HOGS w/ Physical Losses	**1.651**

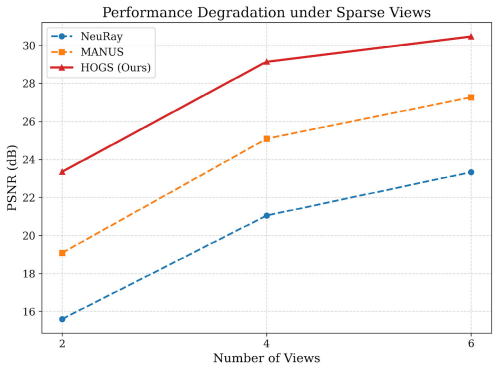


Fig. 12. **Performance degradation curve** under decreasing camera views.

4) Analysis of Physically Plausible Optimization: To quantify the effect of our physical loss, we introduce the Mean Squared Penetration Depth (MSPD) metric. This metric averages the squared penetration depths of human Gaussian means (from predicted contact regions $\mathcal{C}$, see Sec. III-C.1) into the object's SDF. Specifically, MSPD is calculated as $\frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}}(\max(0,-\delta(\mu_i^h)))^2$, where $\delta(\mu_i^h)$ is the signed distance of the i-th human Gaussian mean μ_i^h within the object's SDF. A lower MSPD value indicates less penetration and thus better physical plausibility. As detailed in Table VI, applying our physical losses leads to a significant reduction in MSPD, decreasing the value from 3.829 cm^2 to 1.651 cm^2, which confirms their effectiveness in mitigating interpenetration.

Complementing these quantitative findings, Figure 13 qualitatively demonstrates the impact of physical losses. As shown in Figure 13(i), rendering without physical losses causes the human to float above the sofa, lacking plausible physical

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(i) w/o Physical Loss (ii) w/ Physical Loss

Fig. 13. **Qualitative effect of our physical losses**. (a) Without physical losses, the rendered human floats above the chair, showing a clear lack of physical contact. (b) With our losses, HOGS produces a plausible interaction where the person is sitting correctly on the chair.

TABLE VII

QUANTITATIVE ROBUSTNESS EVALUATION UNDER EXTREME VIEW SPARSITY. NOTE THAT THE RESULTS FOR THE STANDARD SETTING (6 VIEWS) ARE REPORTED IN TABLE I

Method	2 Views		4 Views	
	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
NeuRay [100]	15.60	0.725	21.05	0.854
MANUS [40]	19.08	0.845	25.10	0.912
HOGS (Ours)	**23.36**	**0.887**	**29.15**	**0.942**

contact. Conversely, Figure 13(ii) yields significant visual improvements. The rendered human now exhibits plausible contact with the chair, showcasing a more realistic and physically consistent interaction. This visual evidence, combined with the MSPD results, strongly validates the effectiveness of our proposed physical losses in enforcing plausible HOIs.

D. Further Analysis

1) Robustness to Camera View Sparsity: Existing sparse-view methods often degrade severely when the number of input views drops below a certain threshold. To evaluate the robustness of HOGS under extreme sparsity, we conduct a stress test by reducing the number of input views from 6 to 4 and 2. We compare our method against the leading NeRF-based method (NeuRay [99]) and 3DGS-based method (MANUS [40]).

Table VII presents the results under these extreme settings, while the standard 6-view performance serves as the baseline (refer to Table I). The degradation trends are visualized in Figure 12. As observed, while all methods experience performance drops as views decrease, HOGS exhibits the most graceful degradation. Specifically, at the extreme setting of 2 views, HOGS maintains a PSNR of 23.36 dB, significantly outperforming the 3DGS-based baseline MANUS (19.08 dB) by 4.28 dB. Remarkably, by cross-referencing Table VII with Table I, our performance at 2 views (23.36 dB) matches that of the NeRF-based NeuRay at 6 views (23.34 dB). This resilience stems from our physically plausible optimization: when visual cues are critically lacking, the introduced geometric constraints (contact and separation) effectively prevent the geometric collapse that plagues baseline methods.

TABLE VIII

ROBUSTNESS EVALUATION USING DIFFERENT OBJECT MESH SOURCES. WE COMPARE HOGS AGAINST THE SOTA METHOD MANUS [40] USING BOTH HIGH-FIDELITY TEMPLATES AND COARSE MESHES RECONSTRUCTED FROM SPARSE VIEWS

Method	Object Mesh Source	PSNR ↑	Degradation ↓
MANUS [46]	Pre-scanned Template	27.28	-
MANUS [46]	Reconstructed [105]	24.85	-2.43 dB
HOGS (Ours)	Pre-scanned Template	**30.47**	-
HOGS (Ours)	Reconstructed [105]	**28.71**	**-1.76 dB**



(i) Image (ii) Dataset GT (iii) Predicted Mask

Fig. 14. **Visual comparison of masks.** (i) Original Image. (ii) Dataset Reference exhibits gaps at the contact area. (iii) SAM predicts a continuous mask.

2) Flexibility of Object Source: The object deformation stage (Sec. III-A) aims to acquire an object mesh, which serves as the basis for initializing object Gaussians. To align with NeuralDome [24], we directly employ its provided pre-scanned object templates. However, the requirement for pre-scanned templates is impractical for broader applications.

To assess HOGS's viability under more general conditions, we replace the template-based approach by reconstructing object meshes from sparse-view images using Pixel2Mesh++ [104], thus unifying the input modality for both human and object deformations to sparse RGB images. Using these coarse reconstructed meshes naturally introduces geometric inaccuracies, which can affect the precision of initial Gaussian placement. As shown in Table VIII, this leads to a performance drop for all methods.

However, a comparative analysis highlights the superior robustness of our framework. When forced to use coarse reconstructed geometry, the state-of-the-art method MANUS [40] suffers a severe degradation of 2.43 dB. In contrast, HOGS demonstrates significantly better resilience with a smaller drop of 1.76 dB. This suggests that our physically plausible optimization can better compensate for geometric initialization errors. Crucially, even with these "noisy" reconstructed inputs, HOGS (28.71 dB) still outperforms the "perfect" version of the SOTA baseline (MANUS at 27.28 dB), confirming its practical viability in template-free scenarios.

3) Robustness to Segmentation Mask Quality: In our main experiments, we utilize reference masks provided by the HODome dataset, generated via Background Matting [105]. While accurate, this approach requires a pre-captured clean background, limiting in-the-wild applicability. To evaluate robustness and practical deployment potential, we adopt the Segment Anything Model (SAM) [106] to extract masks using only coarse bounding-box prompts from the current frame.

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TABLE IX

ROBUSTNESS EVALUATION ON SEGMENTATION MASK QUALITY. COMPARISON BETWEEN DATASET REFERENCE (BACKGROUND MATTING) AND SAM-PREDICTED MASKS

Mask Source	Input Requirement	PSNR ↑	SSIM ↑
HODome (Ref.)	Image + Background	30.47	0.949
Predicted (SAM)	Image Only	30.51	0.950

TABLE X

EFFICIENCY COMPARISON AGAINST ADAPTED MESH-BASED CONTACT STRATEGIES AND GLOBAL SEARCH BASELINES. OUR PREDICTION-GUIDED SDF SCHEME ACHIEVES ~ 17× SPEEDUP

Category	Method	Time / Iter. (s) ↓	Speedup ↑
Mesh-Based (Adapted)	BEHAVE [16]	6.259	1.0×
	CHORE [94]	6.527	1.0×
	GraviCap [96]	6.325	1.0×
3DGS-Based	Global Search	5.722	1.1×
	Ours	**0.375**	**17.4**×

TABLE XI

TRAINING AND OPTIMIZATION TIMES. PRE-TRAINING IS A ONE-TIME COST. PER-SCENE OPTIMIZATION IS FOR ADAPTING TO A NEW SEQUENCE

Method / Component	Time (min)
HOGS: Human Pose Refinement (Pre-train)	∼15
HOGS: Contact Prediction (Pre-train)	∼24
HOGS: Main Pipeline (Per-Scene Optim.)	7.5
MANUS [40] (Per-Scene Optim.)	9

We conduct an experiment on the Subject 01 subset by replacing dataset-provided masks entirely with SAM-predicted masks. As shown in Table IX, using SAM-predicted masks yields comparable and slightly superior results to the original dataset reference. The reason for this improvement is visualized in Figure 14: the SAM-predicted mask is more faithful to physical reality, specifically depicting the human and object in full contact (Figure 14(iii)), whereas background matting often leaves artifactual gaps (Figure 14(ii)). Crucially, the performance stability between these two distinct mask sources highlights the effectiveness of our physically plausible optimization. Our geometric constraints (Sec. III-C) act as regularizers, allowing the model to recover plausible contact geometry even when input masks exhibit inconsistencies.

4) Efficiency and Training Time: Our framework is designed for both high-quality rendering and efficiency. As shown in Table XI, the *Sparse-View Human Pose Refinement* and *Sparse-View Human-Object Contact Prediction* modules are pre-trained offline as a one-time cost (∼15 and ∼24 minutes, respectively). Subsequently, for each HOI scene, the main HOGS pipeline performs per-scene optimization.

To validate the computational advantage of our scheme, we compare HOGS against two baselines: (1) **Adapted Mesh-based Methods**, where we implement contact losses from representative works (BEHAVE [16], CHORE [93], GraviCap [95]) using Nearest Neighbor search adapted for Gaussians; and (2) **Global Search**, an ablation of our method that computes SDF queries for all Gaussians without contact prediction.

As shown in Table X, the adapted mesh-based methods are computationally expensive (∼6.3s per iteration) due to the complexity of neighbor searching in unstructured Gaussian fields. Even using efficient SDF queries (Global Search) only reduces this to 5.7s. In contrast, by combining SDF queries with our predicted contact mask, HOGS narrows the optimization scope significantly, reducing the physics calculation to just **0.375s per iteration**. This achieves a speedup of over 16× compared to standard approaches, confirming that our design is critical for enabling real-time, physically plausible optimization.

This efficiency is critical for the overall pipeline. Thanks to this accelerated physics computation, HOGS's main pipeline requires only 7.5 minutes of total optimization time per scene. This is notably more efficient than the baseline MANUS [40], which requires 9 minutes. This highlights HOGS's advantage in rapid adaptation for novel HOI sequences.

V. CONCLUSION

In this paper, we present HOGS, a novel approach for rendering realistic and physically plausible HOIs from sparse views. Its key innovation is the integration of composed Gaussian Splatting with physically plausible rendering optimization, supported by sparse-view human pose refinement and human-object contact prediction. This combination enables HOGS to effectively capture intricate human-object interplay, generating high-quality novel views that adhere to physical constraints. Extensive experiments demonstrate HOGS achieves superior performance in rendering quality, efficiency, and physical plausibility over existing methods, while also showing broader applicability to articulated object interactions.

Limitations and Future Work. While HOGS marks a significant step forward for HOI rendering, the current framework mainly focuses on interactions between a single human and a single object. This is a common setup in foundational HOI research, but extending it to more complex scenes is a crucial next step. We consider that our component-based design provides a clear path for such extensions. One could extend HOGS to multi-person or multi-object scenarios by creating multiple instances of our human/object deformation module and introducing inter-entity physics (e.g., human-human or object-object collision) into the optimization. Furthermore, to handle non-rigid objects, the current ICP-based object deformation module could be replaced with more advanced models for articulated or deformable objects, while keeping the core physically plausible rendering pipeline intact.

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